



FINAL BUSINESS INSIGHTS REPORT — PRIMEOR SOLUTIONS

Data Analytics Internship Program

Global E-Commerce Sales & Profitability Analysis (2011–2014)

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Abstract

This report combines the analysis performed during the one-month Data Analytics Internship at Primeor Solutions. It brings together data cleaning and preparation performed using Excel, SQL-based analysis, and an interactive Power BI dashboard into a single narrative covering sales performance, key business trends, profitability, performance drivers, customer behaviour and areas for improvement across the global e-commerce dataset from 2011-2014, providing actionable insights into data-driven decision-making.

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01 Executive Summary

Over the course of the internship, a global e-commerce transaction dataset spanning **2011 to 2014** was cleaned, structured and analyzed to uncover the drivers of sales and profitability across markets, regions, product categories and sub-categories, and customer segments. The workflow contained Excel-based data cleaning, basic analysis, followed by SQL analysis for aggregation and ranking, and finally, Power BI for interactive visualization of the insights summarized in this report.

\$12.63M Total Sales	\$1.47M Total Profit	11.62% Overall Profit Margin
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51K Total Orders	796 Unique Customers	147 Countries Served
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Key findings: the **APAC** market generated the highest total revenue among all regions analyzed, and the **Central** region led in sales within the dataset's region field. **Technology** products delivered by far the strongest average profitability, while **Office Supplies** lagged behind. **Standard Class** shipping dominated fulfillment, used in roughly six out of every ten orders, and **Furniture** products carried the steepest average discounting — a likely contributor to their comparatively thin profit margins (~7%).

02 Project Scope & Methodology

The internship followed a structured, end-to-end analytics workflow, moving raw transactional data through cleaning, transformation, querying and visualization to arrive at decision-ready insights.

1. Data Preparation (Excel)

The raw Global E-commerce dataset (51,272 items) was inspected for missing values, inconsistent formats and duplicate records (54 records were eliminated), incorrect date formats and invalid numerical fields, then structured into a clean, analysis-ready table covering order, customer, product, geography and financial fields.

2. SQL Analysis

A dedicated MySQL database (primeor_analysis_queries) was built from the cleaned dataset. Date fields were converted to proper DATE types, financial columns were converted to DECIMAL precision, and a series of aggregation, ranking (RANK() OVER) and queries were written to surface top products, top customers, regional and market performance, monthly trends and shipping mode usage.

3. Interactive Dashboard (Power BI)

Key metrics and dimensions were modeled into an interactive Power BI dashboard (Primeor_DAInt_Ecom_Analysis_Dashboard.pbix), enabling drill-down exploration of sales, profit, discount and shipping patterns across time, geography and product hierarchy.

4. Insight Synthesis

Results from the SQL analysis and dashboard were cross-validated and synthesized into the key business insights and recommendations presented in this report.

Tools used: Microsoft Excel · MySQL · Power BI

Deliverables submitted: Cleaned Dataset (.csv), SQL Queries File (.sql), Excel Analysis File (.xlsx), SQL results and insights (.xlsx), Power BI Dashboard (.pbix), and this Final Business Insights Report (.pdf).

03 Key Business Insights

The six insights below represent the most actionable findings from the SQL analysis and dashboard exploration.

1

APAC leads market revenue; Central leads by region

APAC generated the highest total revenue of any market (\$3.58M), narrowly ahead of the EU (\$2.94M). Within the dataset's regional breakdown, Central was the top-selling region at \$2.82M in sales.

2

Standard Class dominates order fulfillment

Standard Class was used in roughly 60% of all shipments, more than the other three shipping modes combined, which sends a strong signal of cost-conscious, non-urgent fulfillment behaviour across the global customer base.

3

Technology is the most profitable; Office Supplies lags

Technology products delivered the highest average profit per order (\$65.46), roughly 2.3x that of Furniture (\$29.00) and nearly 4x that of Office Supplies (\$16.58). The top-10 most profitable products table is dominated by Technology items, further supporting this pattern.

4

Tom Ashbrook is the top customer by sales

Tom Ashbrook ranks first among all customers by total sales (\$40,489), narrowly ahead of Tamara Chand (\$37,453) and Greg Tran (\$35,552), highlighting the value concentrated in a small set of high-spending accounts. However, when considering **individual purchases**, Sean Miller recorded the highest single expenditure, amounting to \$23,481.

5

Furniture carries the heaviest discounting

Furniture had the highest average discount rate of any category at 17%, and the single deepest discount in the dataset (85%) was applied to a Furniture item — a plausible driver of the category's comparatively modest profitability, showing a profit margin of only about 7%.

6

Phones are the top-performing sub-category

Phones generated the highest sales of any sub-category (\$1.71M), narrowly ahead of Copiers (\$1.51M) and Chairs (\$1.50M), making them the single most important product line by revenue.

04 Market & Regional Performance

Revenue is heavily concentrated in a handful of markets. APAC and the EU together account for more than half of total company revenue, while Canada contributes a negligible share — suggesting either a very small footprint or an early-stage market in that region.

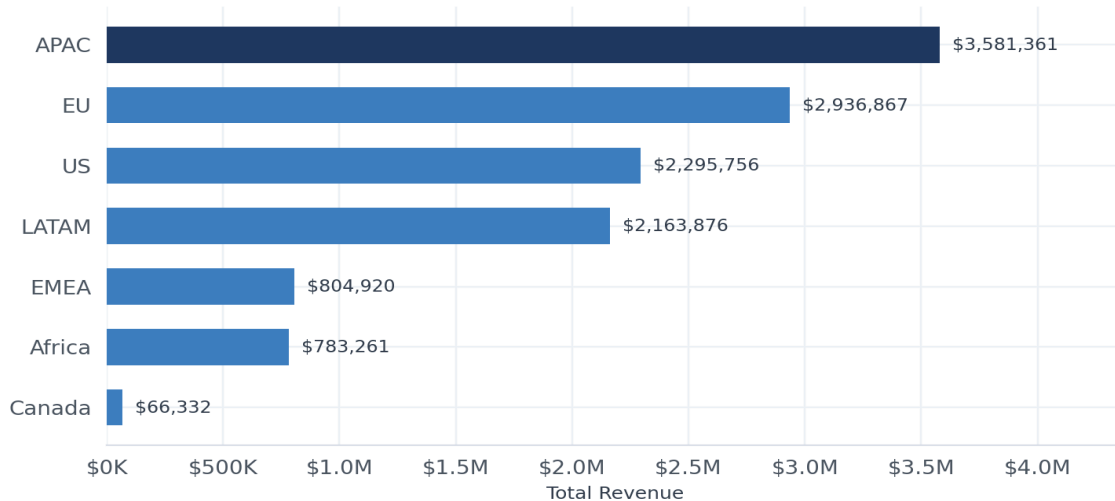


Figure 4.1 — Total revenue by market

Looking at the **region** field, Central, South and North lead the set, while smaller regions such as the Caribbean and Canada trail well behind the top performers — consistent with the market-level sales performance seen above.

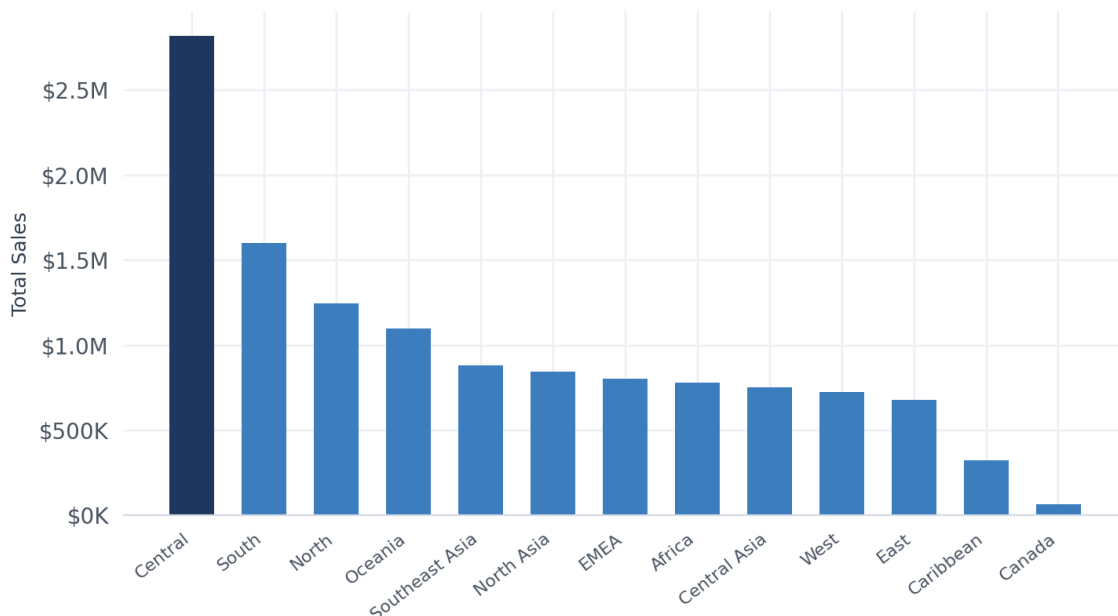


Figure 4.2 — Total sales by region

05 Sales Trend Analysis

Aggregating sales by calendar month across all four years reveals a clear seasonal pattern: sales build steadily through the year and peak in the final quarter, with November and December the two strongest months. This points to a strong seasonal/holiday-driven demand cycle worth planning inventory and marketing spend around. Although right after that, February witnesses the lowest sales performance every year.

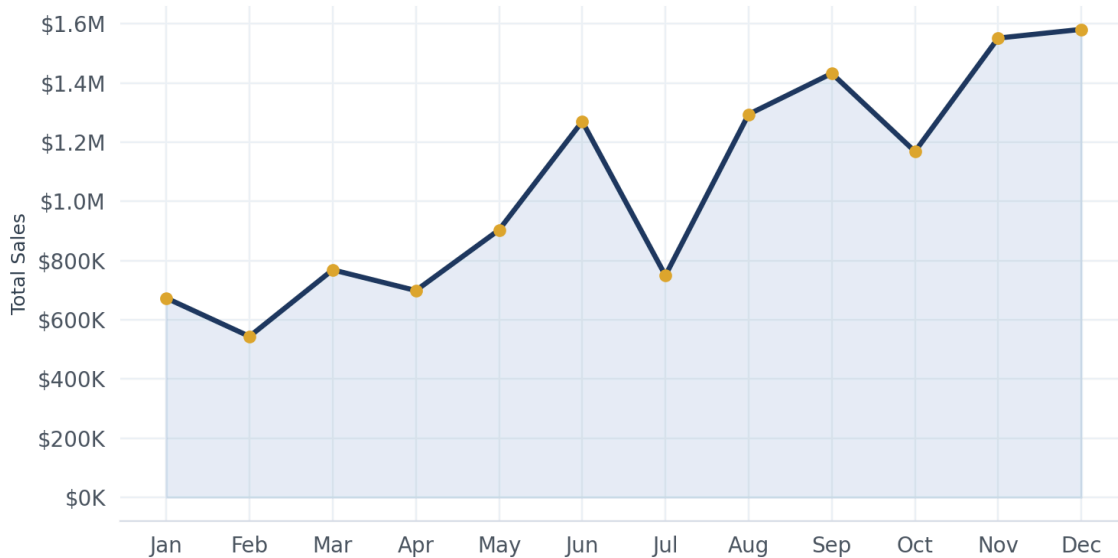


Figure 5.1 — Total sales by calendar month (aggregated across all years)

At the year level, total sales grew consistently year over year across the 2011–2014 window, reflecting steady business expansion over the period covered by the dataset.



Figure 5.2 — Total sales by year

06 Product & Category Performance

The Canon imageCLASS 2200 Advanced Copier is the single most profitable product in the dataset, generating over \$25,000 in total profit — nearly 50% more than the next best performer. Technology products occupy 7 of the top 10 spots, underscoring the category's outsized contribution to bottom-line profitability.

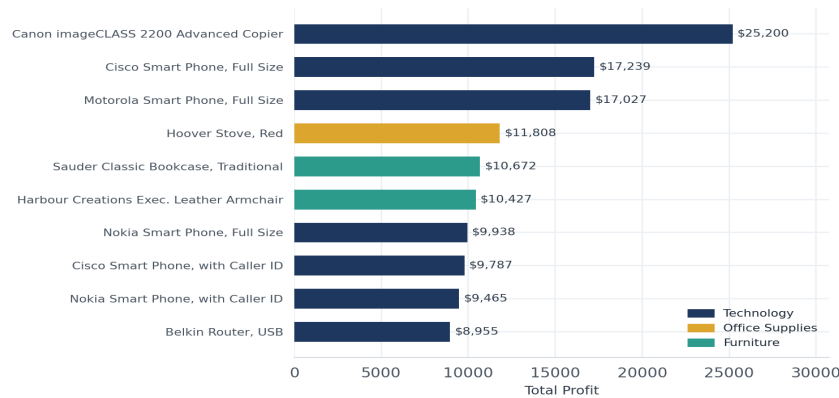


Figure 6.1 — Top 10 products by total profit

Average profit per order differs sharply by category. Technology's average profit per order is more than double that of Furniture and nearly four times that of Office Supplies, making it the clear priority category for continued investment and promotion.



Figure 6.2 — Average profit by category

At the sub-category level, Phones, Copiers and Chairs are the three strongest revenue generators, together contributing roughly \$4.7M — more than a third of total company sales.

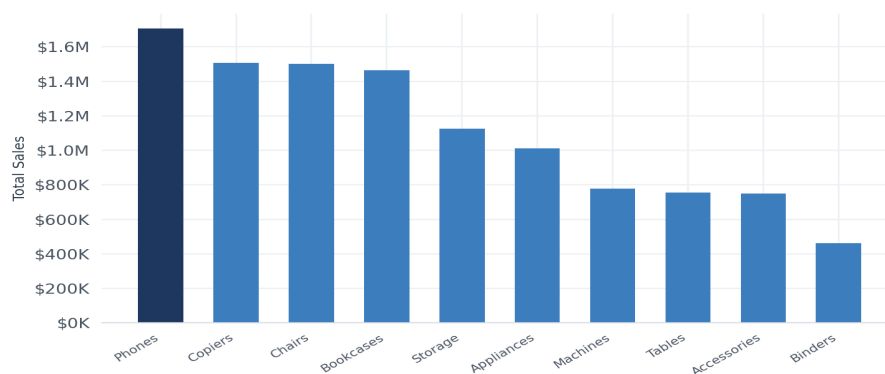


Figure 6.3 — Total sales by top 10 sub-categories

07 Customer Insights

The top 10 customers by total sales are a mix of high-value repeat buyers whose combined spend significantly exceeds the average customer. These accounts can be prime targets of upsell campaigns.

Rank	Customer Name	Total Sales
1	Tom Ashbrook	\$40,489
2	Tamara Chand	\$37,453
3	Greg Tran	\$35,552
4	Christopher Conant	\$35,187
5	Sean Miller	\$35,131
6	Bart Watters	\$32,315
7	Natalie Fritzler	\$31,778
8	Fred Hopkins	\$30,404
9	Jane Waco	\$30,288
10	Hunter Lopez	\$30,246

Customer Segment Mix

The Consumer segment is the largest source of revenue, generating over half of total sales, followed by Corporate and Home Office. This suggests marketing and retention efforts should continue prioritizing the Consumer segment while still nurturing the higher order-value Corporate accounts.

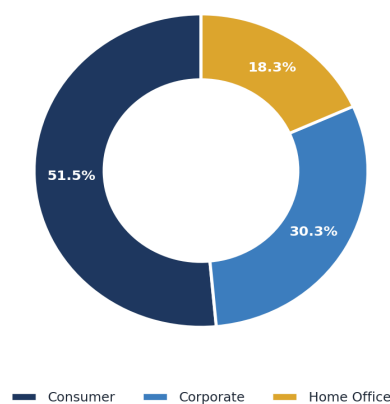


Figure 7.1 — Share of total sales by customer segment

08 Shipping & Operations

Standard Class is by far the most-used shipping mode, accounting for approximately 60% of all orders. Second Class and First Class follow at roughly 20% and 15% respectively, with Same Day delivery used in only about 5% of orders. This distribution suggests most customers prioritize cost over speed, which has implications for logistics capacity planning and any premium-shipping upsell strategy.

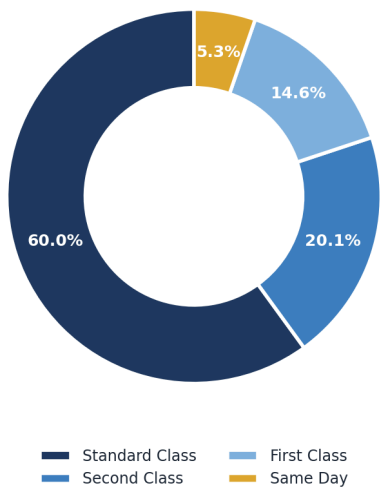


Figure 8.1 — Share of orders by shipping mode

Shipping Mode	Order Count	% of Total
Standard Class	30,756	60.0%
Second Class	10,301	20.1%
First Class	7,503	14.6%
Same Day	2,697	5.3%

09 Data Quality Notes: Loss-Making Orders

The SQL analysis identified a substantial number of individual line items with negative profit (over 1,000 records), ranging from minor losses of a few cents to more significant markdown-driven losses. This could be due to discounting and/or per-unit shipping costs occasionally outweighing the revenue captured.

Recommendation: a follow-up deep dive into the discount thresholds and shipping cost allocation tied to these loss-making orders would help quantify their aggregate drag on overall margin and identify whether specific products, regions or customers are systematically associated with them.

10 Recommendations

1. Concentrate more on highly profitable Technology

Prioritize marketing spend, inventory availability and sales incentives on Technology products — particularly Phones and Copiers — given their massive contribution to profit.

2. Review Furniture discounting

Investigate discount rates for Furniture, where a 17% average discount (and discounts as steep as 85% on individual items) is likely compressing margins.

3. Invest in APAC and EU

Continue building sales capacity in APAC and the EU, the two highest-revenue markets, while running a focused diagnostic on Canada to understand its minimal contribution.

4. Plan for Q4 seasonality

Align inventory, staffing and marketing calendars with the clear November–December sales peak identified in the monthly trend analysis, while investigating whether any particular factor is compressing sales in February every year so that it can be checked.

5. Nurture top accounts

Formalize a key-account program around the top 10 customers by sales, who collectively out-spend the average customer by a wide margin.

11 Conclusion

This internship provided end-to-end, hands-on experience across the full analytics pipeline — from cleaning a large, multi-year transactional dataset in Excel, through building and querying a relational database in SQL, to designing an interactive Power BI dashboard for stakeholder exploration. The resulting analysis surfaced clear, actionable patterns in revenue concentration, product profitability, customer value and shipping behaviour that can be used to guide category strategy, discounting policy and seasonal planning.

All supporting files — the cleaned dataset, SQL query file, Excel analysis workbook, SQL results and insights, and Power BI dashboard — accompany this report and provide full traceability from raw data to the insights and recommendations presented above.

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